

The atomic direct sum of the selected CAR representations

`MathlibAnnex.CStarAlgebra.CAR.selectedAtomicRepresentation`

def

All selected pure-GNS fibers of the completed CAR algebra act together on one Hilbert direct sum.

Statement

Let A be the completed CAR algebra, the norm completion of the unital matrix system $M_{2^n}(\mathbb{C})$ with embeddings $a \mapsto a \otimes I_2$. Let ω be its distinguished pure product-vector state. Index pure-state GNS equivalence classes by I , choose one representative φ_j in each class with $\varphi_o = \omega$ at $o = [\omega]$, and denote its GNS data by (H_j, π_j, ξ_j) . Write $\mathcal{H} = \bigoplus_{j \in I} H_j$ for the Hilbert direct sum, $e_j : H_j \rightarrow \mathcal{H}$ for coordinate inclusion, and $\pi(a)(x_j)_j = (\pi_j(a)x_j)_j$. Here the sum consists of square-summable families and has no countability restriction on I . The selected atomic representation is this coordinatewise unital representation $\pi : A \rightarrow B(\mathcal{H})$.

Definition

The common estimate $\|\pi_j(a)\| \leq \|a\|$ bounds every coordinate action uniformly in j . Hence the coordinatewise formula defines a bounded operator on the square-summable families, with $\|\pi(a)\| \leq \|a\|$. Multiplication, the unit and complex linearity hold coordinatewise. The inner product on the direct sum shows that taking adjoints also holds coordinatewise, giving $\pi(a^*) = \pi(a)^*$. The existing arbitrary-index atomic construction packages these properties.

Assumptions

The completed CAR algebra and its root pure state are fixed. Every fiber is the complete GNS Hilbert space of the chosen state. The index set need not be countable, and the Hilbert direct sum is not assumed separable.

Conclusion

For every $a \in A$, $x \in \mathcal{H}$ and $j \in I$, the defining formula is $(\pi(a)x)_j = \pi_j(a)x_j$. In particular, $\pi(a)e_jx = e_j\pi_j(a)x$ for $x \in H_j$. Thus the selected individual actions are the coordinate restrictions of one displayed representation.

The symbol π_j denotes one fiber action and π denotes their direct sum. The cited faithfulness theorem is a separate property of this CAR representation, obtained from its faithful root summand; faithfulness is not a new input or a field added to this definition.

Main citations

- Definition and its exact construction —
`MathlibAnnex.CStarAlgebra.CAR.selectedAtomicRepresentation`

- The selected individual GNS action —
MathlibAnnex.CStarAlgebra.PureState.selectedRepresentation
- The Hilbert direct sum of selected fibers —
MathlibAnnex.CStarAlgebra.PureState.SelectedAtomicHilbert
- Coordinate inclusion into that Hilbert direct sum —
MathlibAnnex.CStarAlgebra.PureState.selectedEmbedding
- Uniformly bounded coordinatewise operator — MathlibAnnex.Analysis.CStarAlgebra.atomicAction
- The arbitrary-index star representation —
MathlibAnnex.Analysis.CStarAlgebra.atomicRepresentation
- Coordinate formula for the atomic action —
MathlibAnnex.Analysis.CStarAlgebra.atomicRepresentation_apply
- Separate faithfulness of the selected CAR sum —
MathlibAnnex.CStarAlgebra.CAR.selectedAtomicRepresentation_injective

Supporting route explanation

Lean source signature (exact)

```
noncomputable def selectedAtomicRepresentation :
  Representation Limit
    (MathlibAnnex.CStarAlgebra.PureState.SelectedAtomicHilbert completedRootPureState) :=
atomicRepresentation
  (MathlibAnnex.CStarAlgebra.PureState.selectedRepresentation completedRootPureState)
```

In the source	Mathematical meaning
Limit; completedRootPureState	The completed CAR algebra A and its root state ω , bundled with the proved purity property.
PureState.selectedRepresentation completedRootPureState	The family $\pi_j : A \rightarrow B(H_j)$ of GNS actions of one selected representative pure state in every GNS class $j \in I$, based at ω .
PureState.SelectedAtomicHilbert completedRootPureState	The Hilbert direct sum $\mathcal{H} = \bigoplus_{j \in I} H_j$ of square-summable families $(x_j)_j$. The index set I is arbitrary, not required to be countable.
atomicRepresentation (...)	The entire RHS assembles the given family into the coordinatewise action $\pi(a)(x_j)_j = (\pi_j(a)x_j)_j$, a unital complex-linear star representation $A \rightarrow B(\mathcal{H})$.
selectedAtomicRepresentation	This is the direct-sum representation π , distinct from an individual π_j . Individual irreducibility does not claim irreducibility of this direct sum; its faithfulness is a separate theorem.

Further source notes: Here Limit is A and completedRootPureState is ω bundled with its purity proof. Coordinate inclusions e_j are the separately defined selectedEmbedding; they are used to state the coordinate restriction formula.

Earlier published Card and PDF

▼ **Exact content identity**

Declaration: `MathlibAnnex.CStarAlgebra.CAR.selectedAtomicRepresentation`

Accepted content SHA-256: 31d69adcb21f8c0d67d242772866f455d6e39cfe57d7947cd0b2f4a41b67bba8

Accepted source guide SHA-256: 9d255e6f5ab0b8a82903766867c6d72498e219ae2d57756bf7a15ee564337e5f

Source commit: 437e6e46228bbb8e91211ebded349d7a30020e73